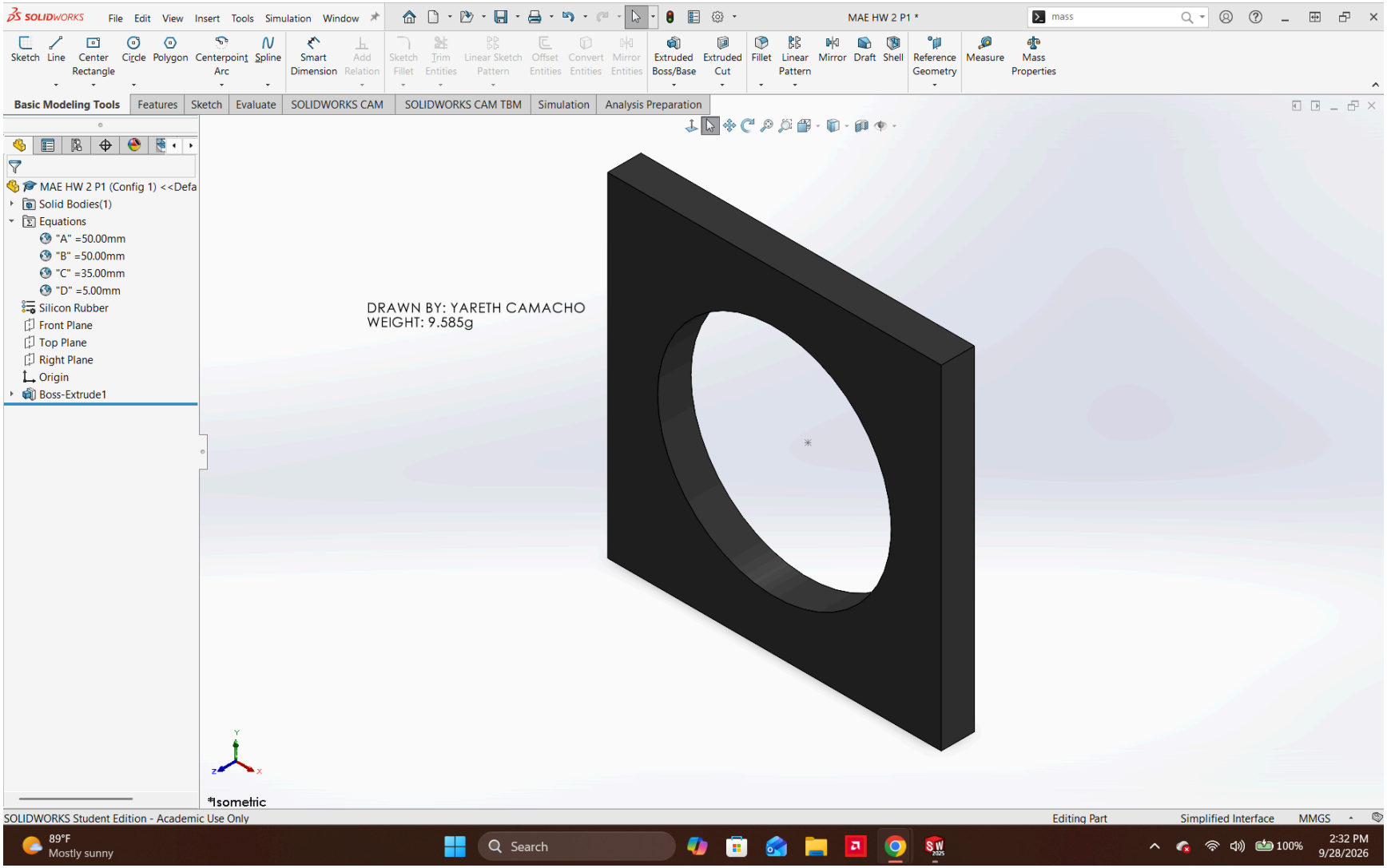
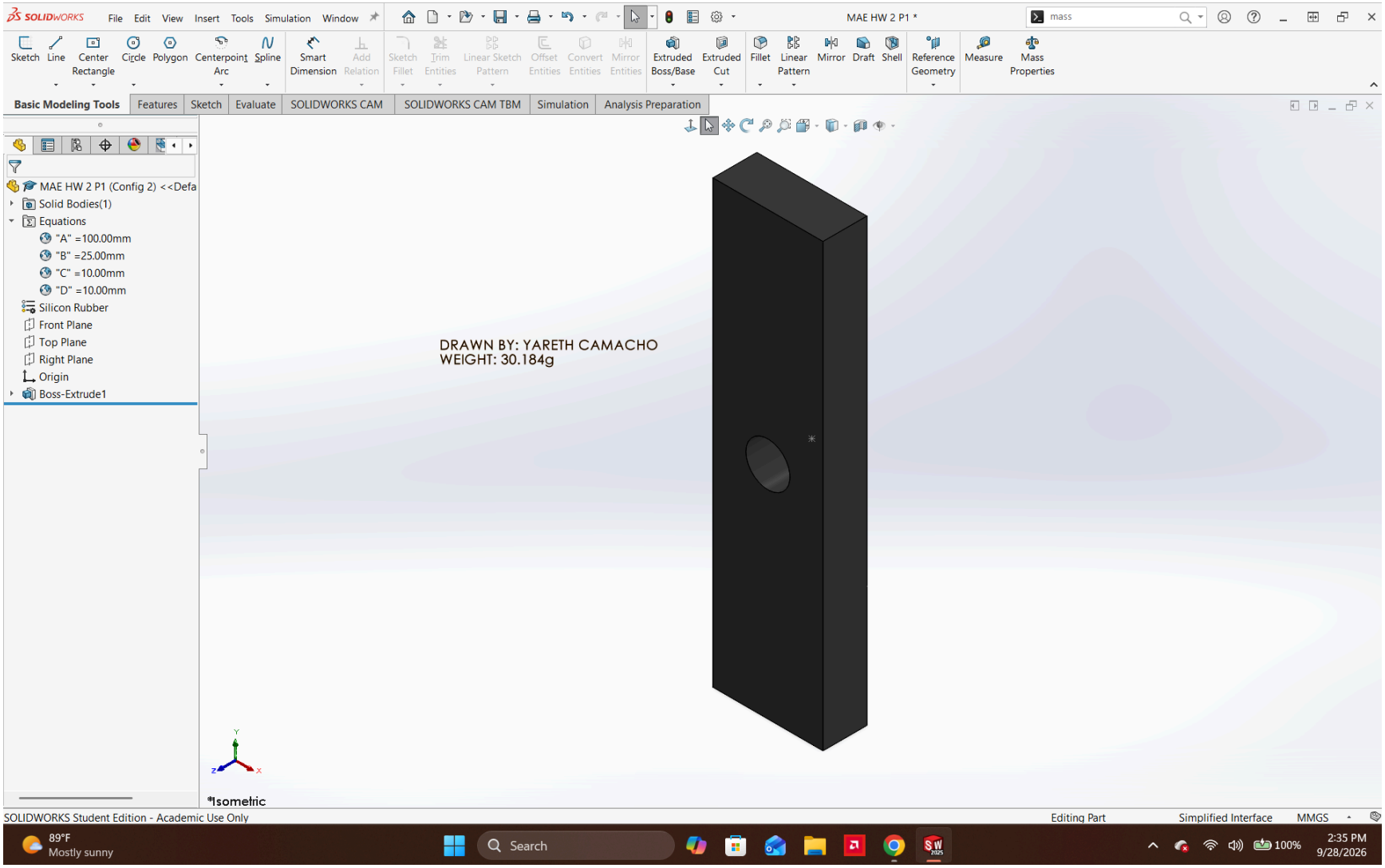
MAE-1351 Homework 2 - Yareth Antonio Camacho

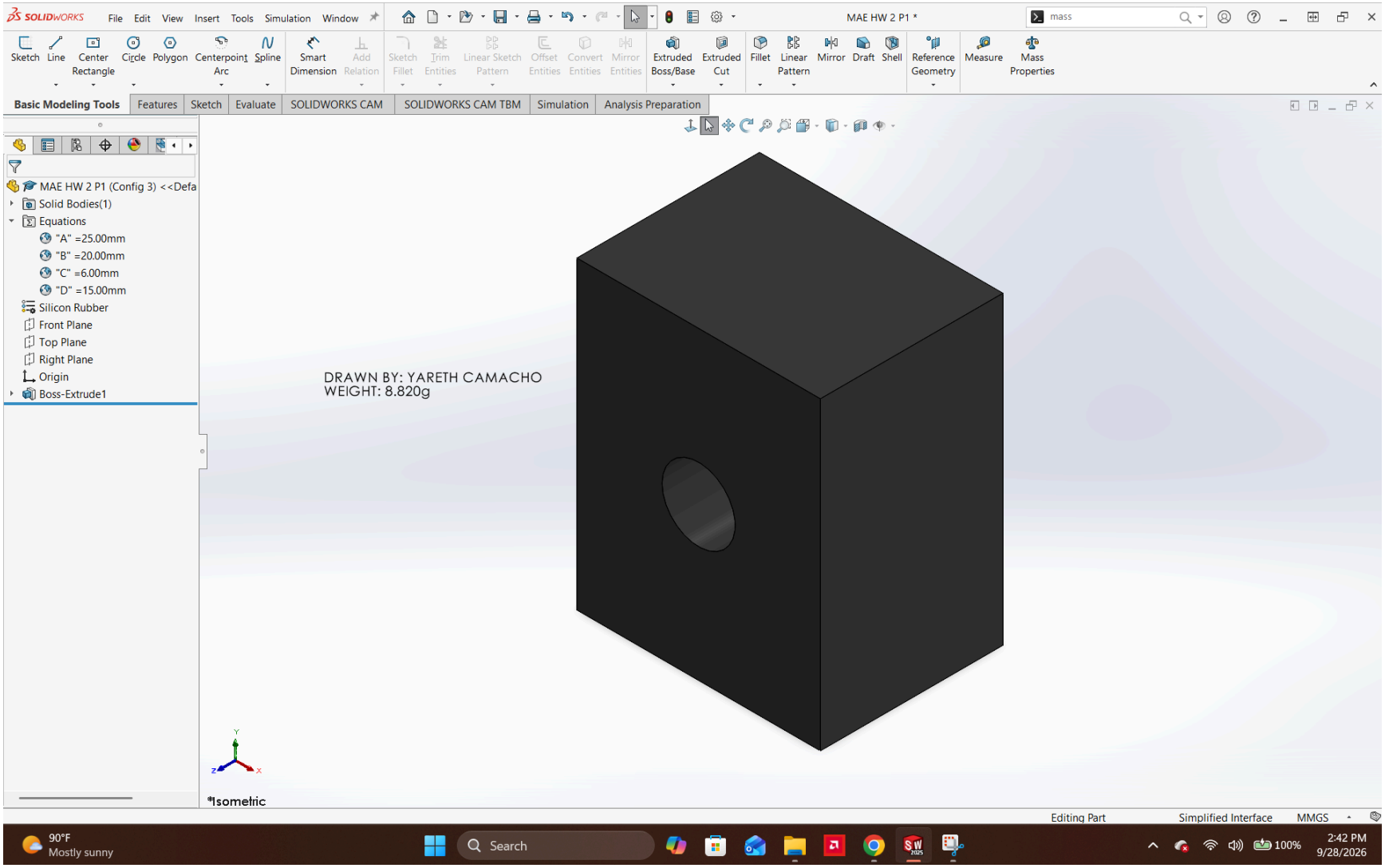
Homework Assignment 1.1



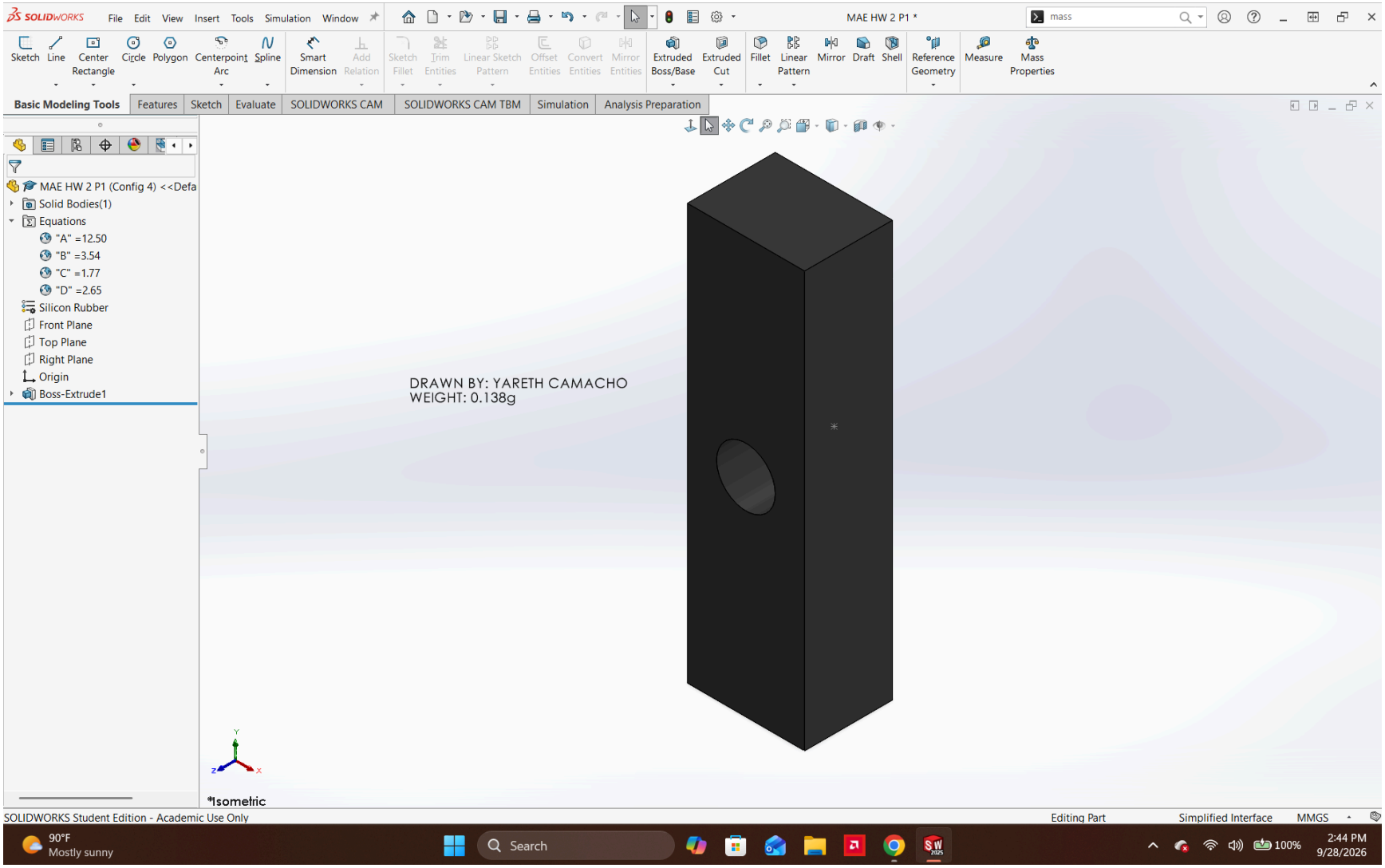
Homework Assignment 1.2



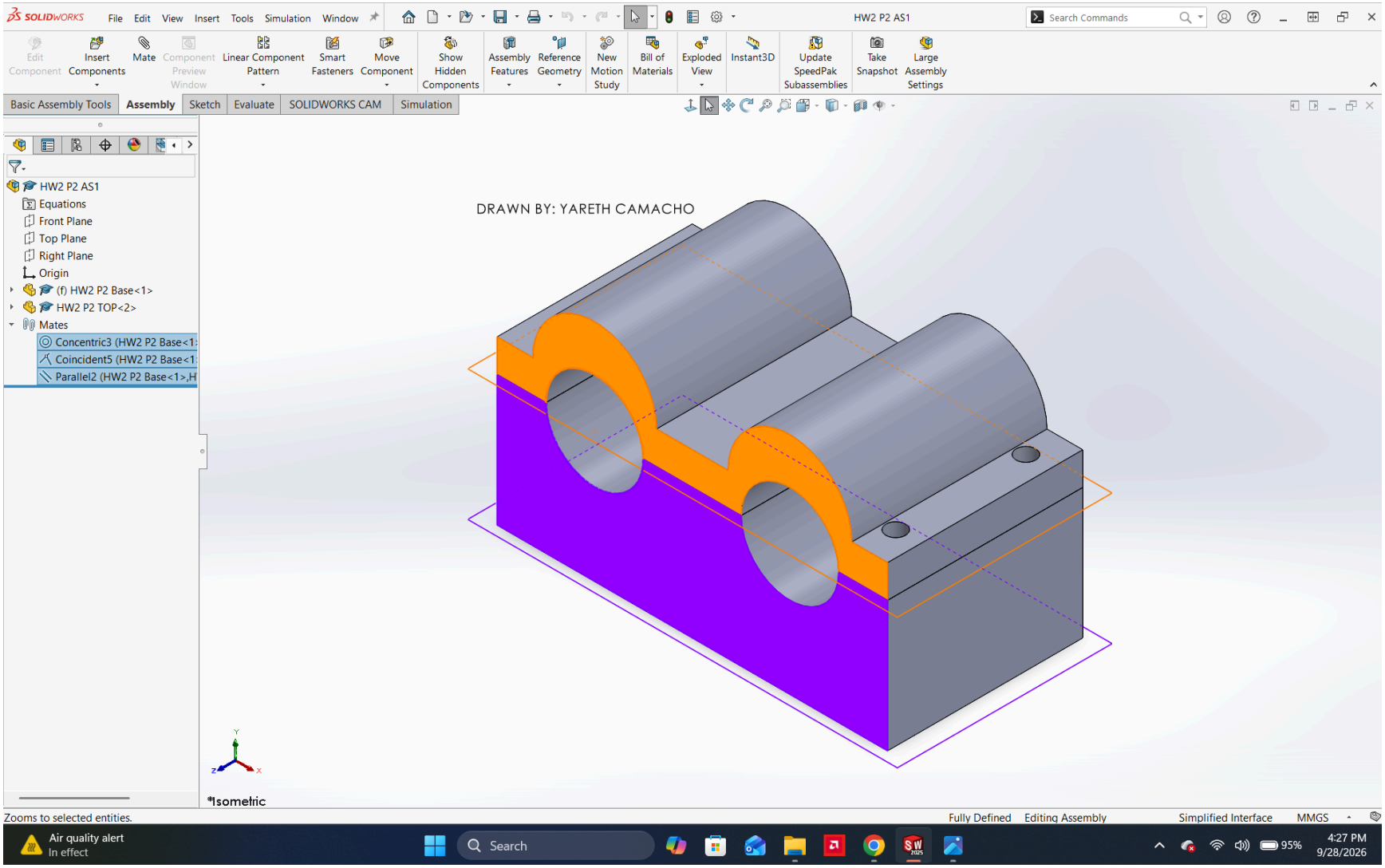
Homework Assignment 1.3



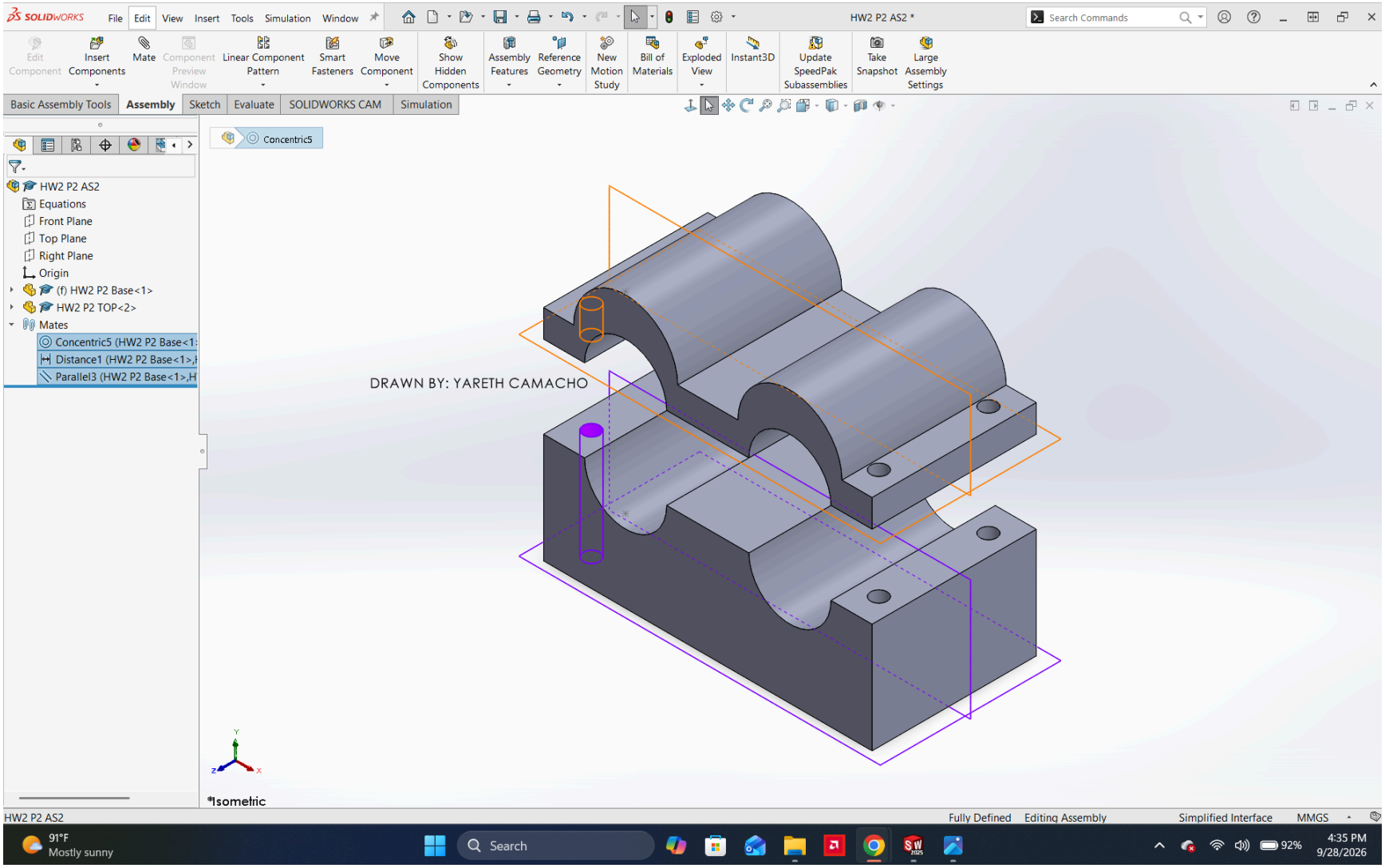
Homework Assignment 1.4



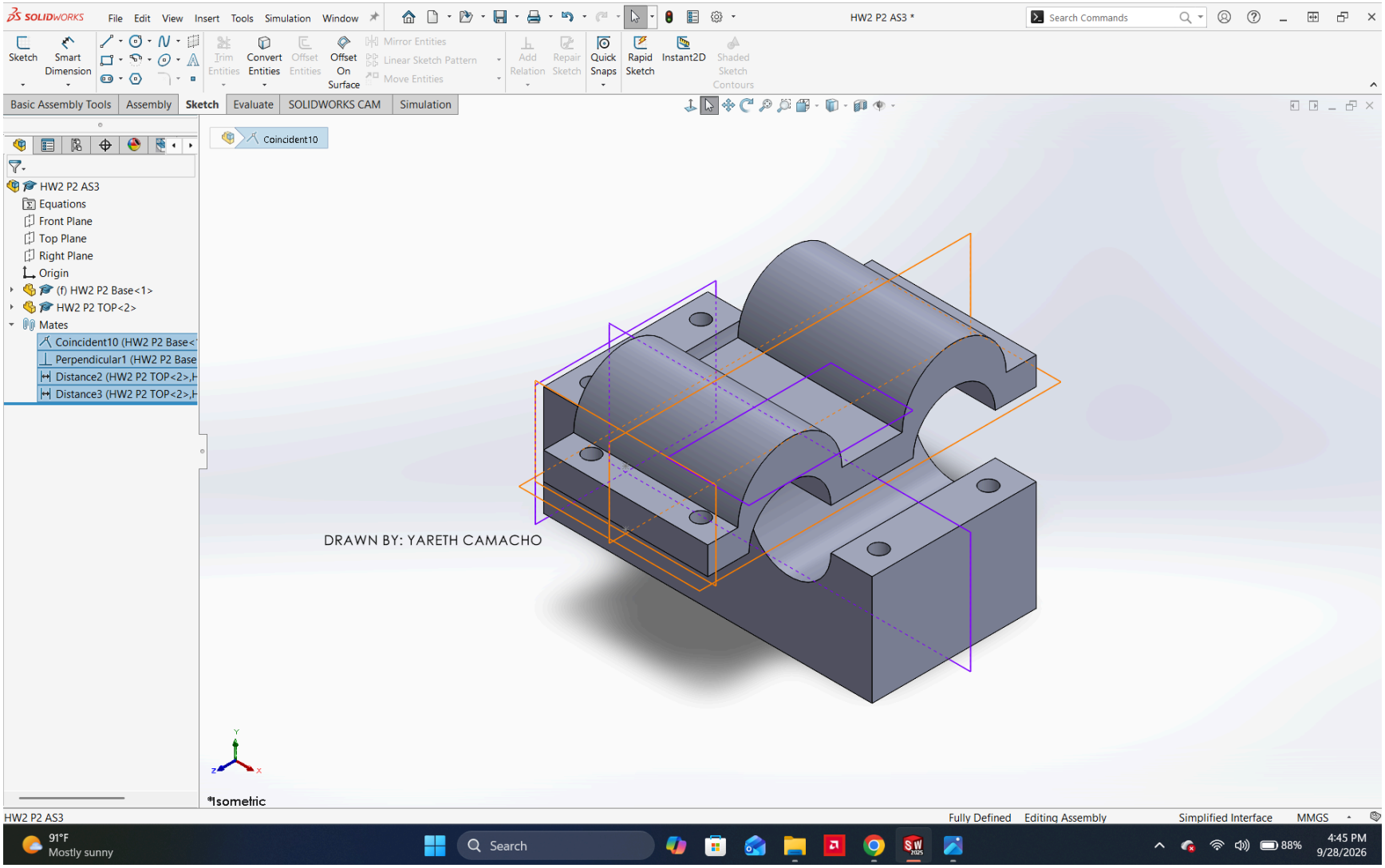
Homework Assignment 2 Assembly 1



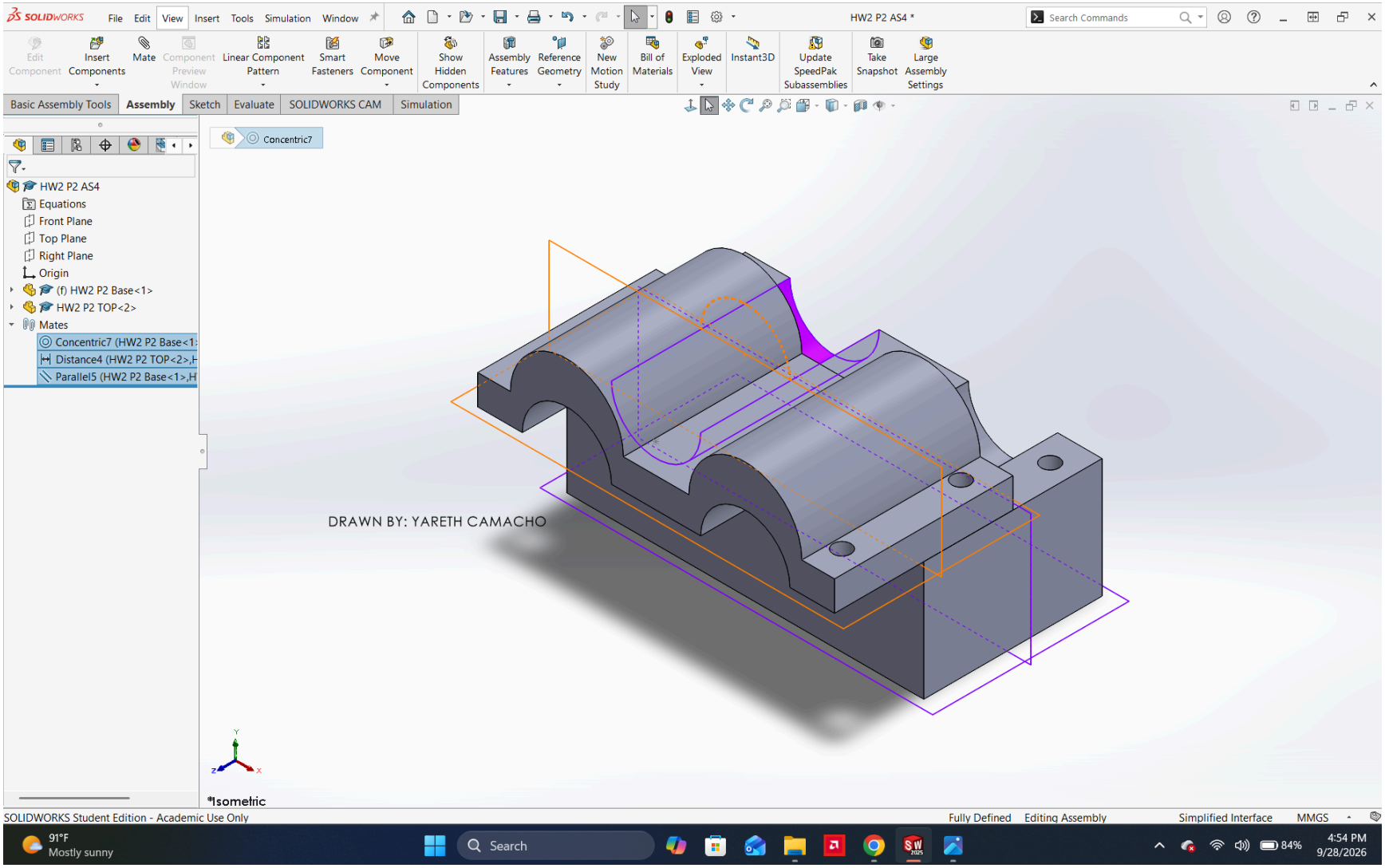
Homework Assignment 2 Assembly 2



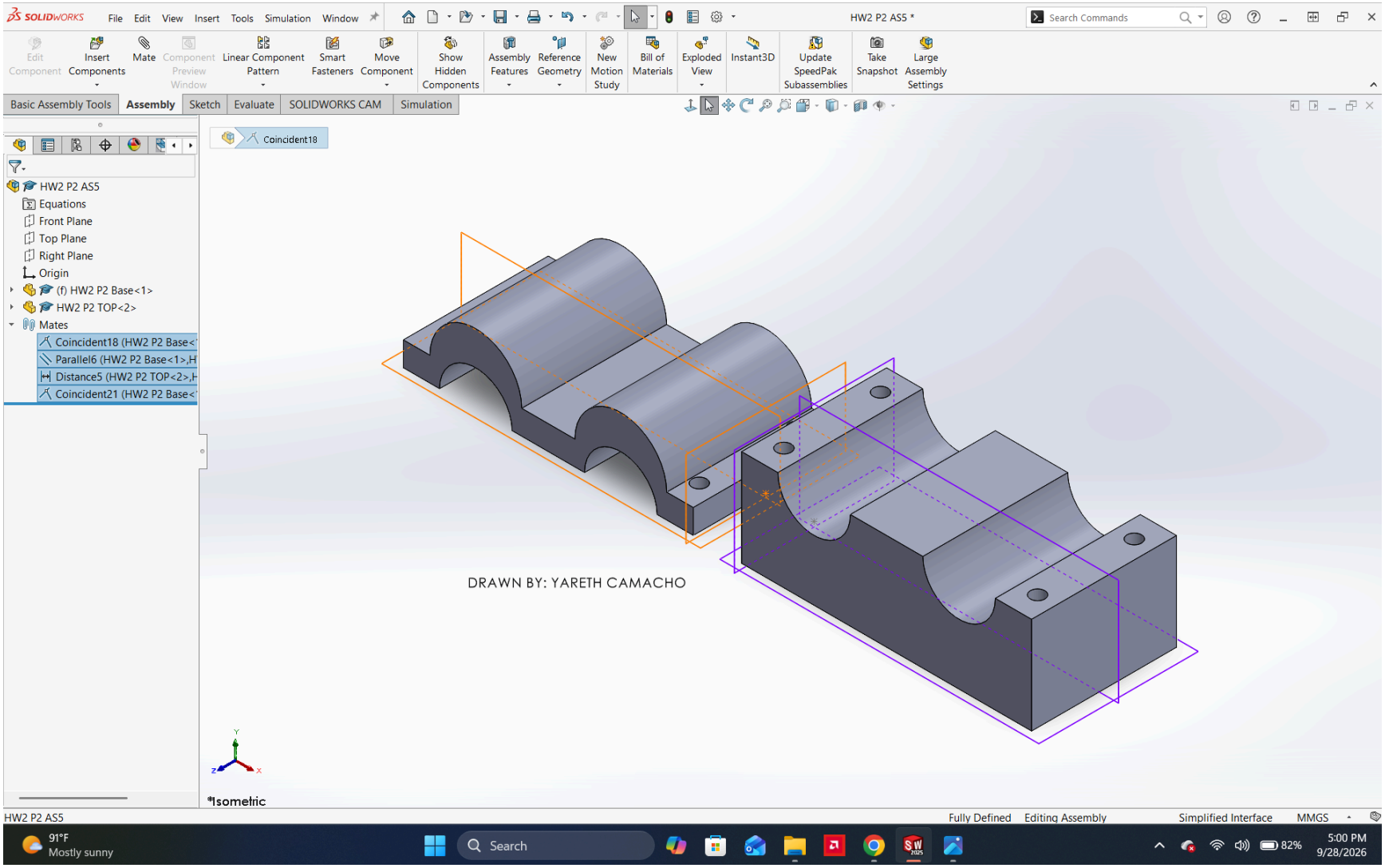
Homework Assignment 2 Assembly 3



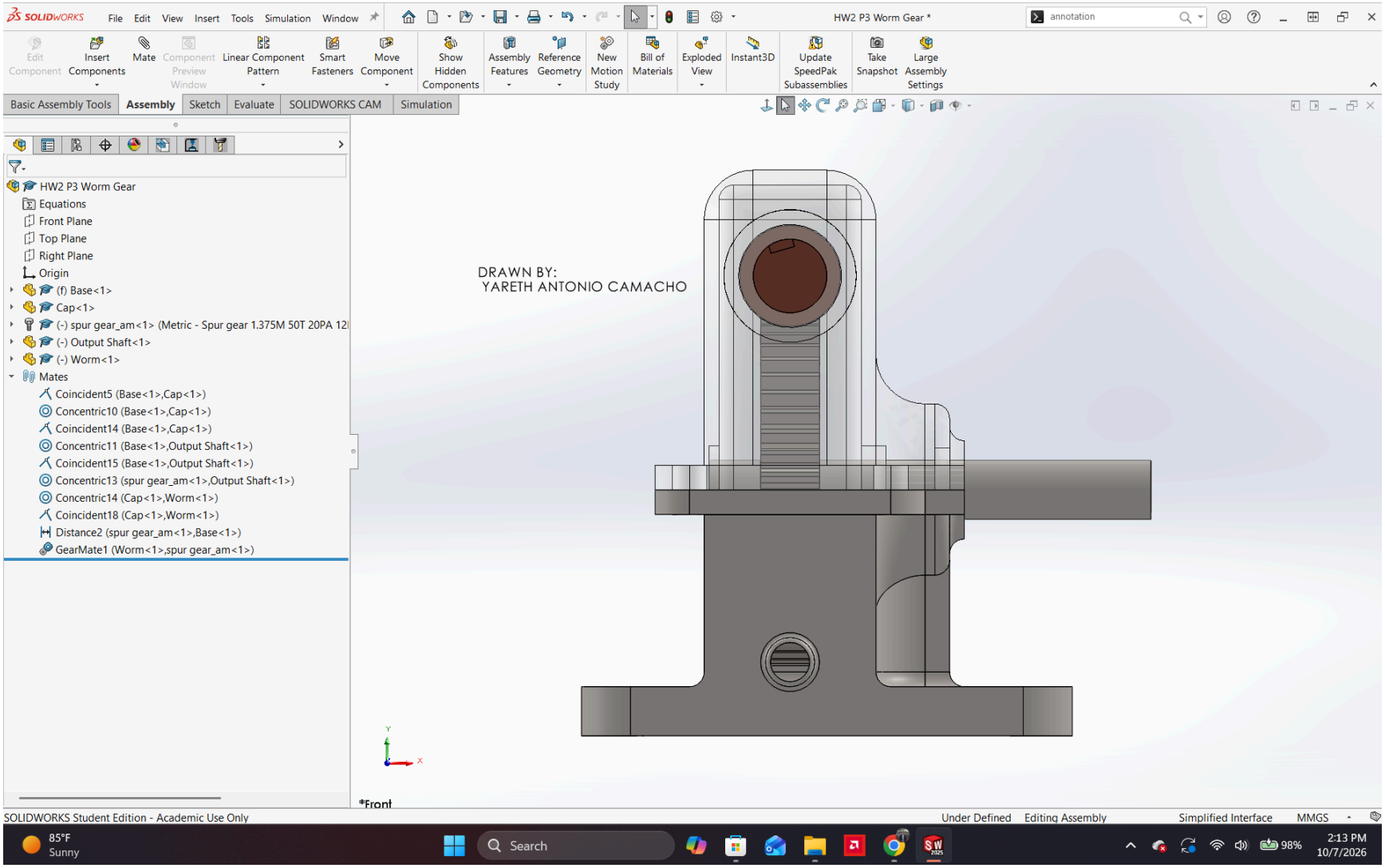
Homework Assignment 2 Assembly 4



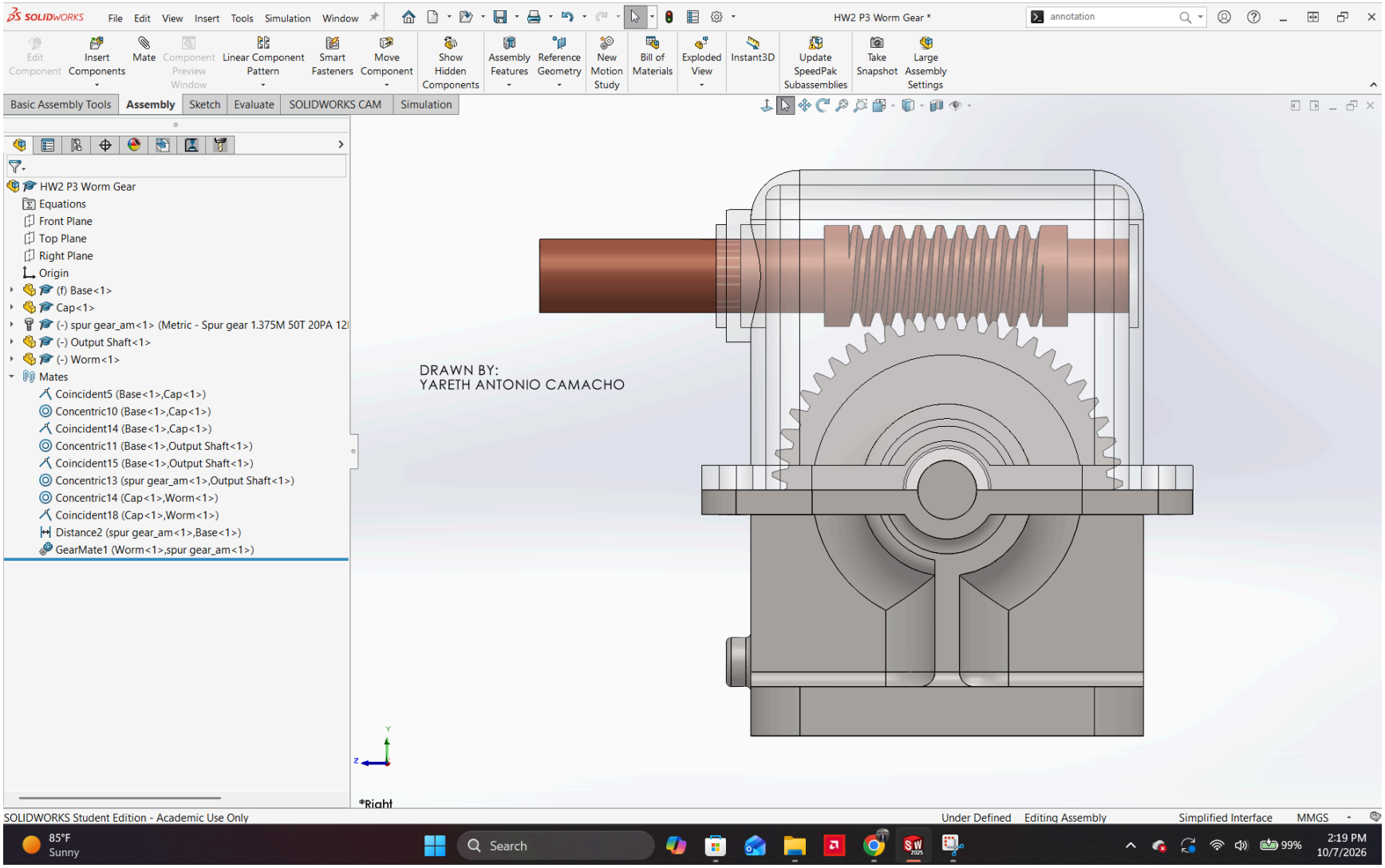
Homework Assignment 2 Assembly 5



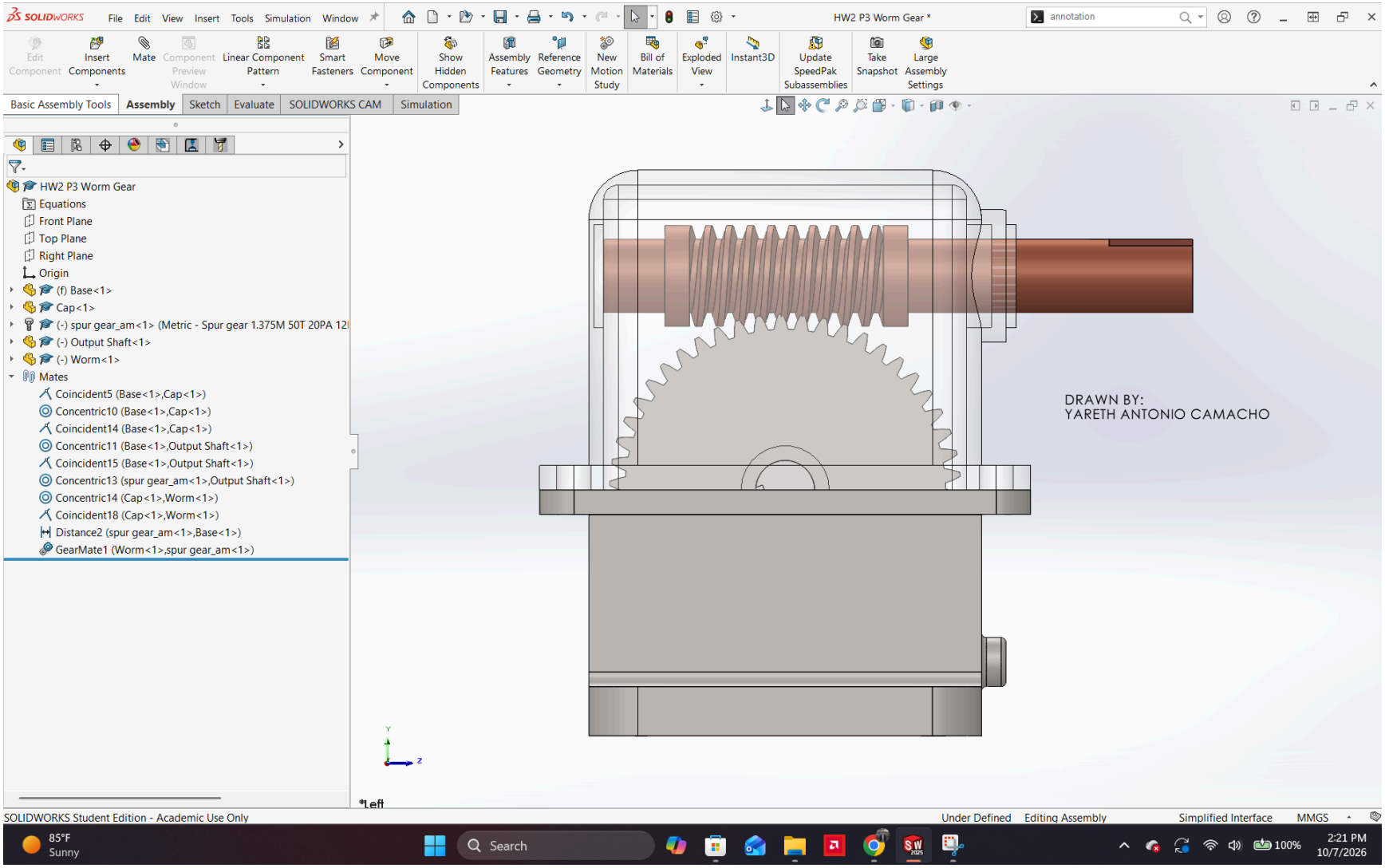
Homework Assignment 3 Assembly 1



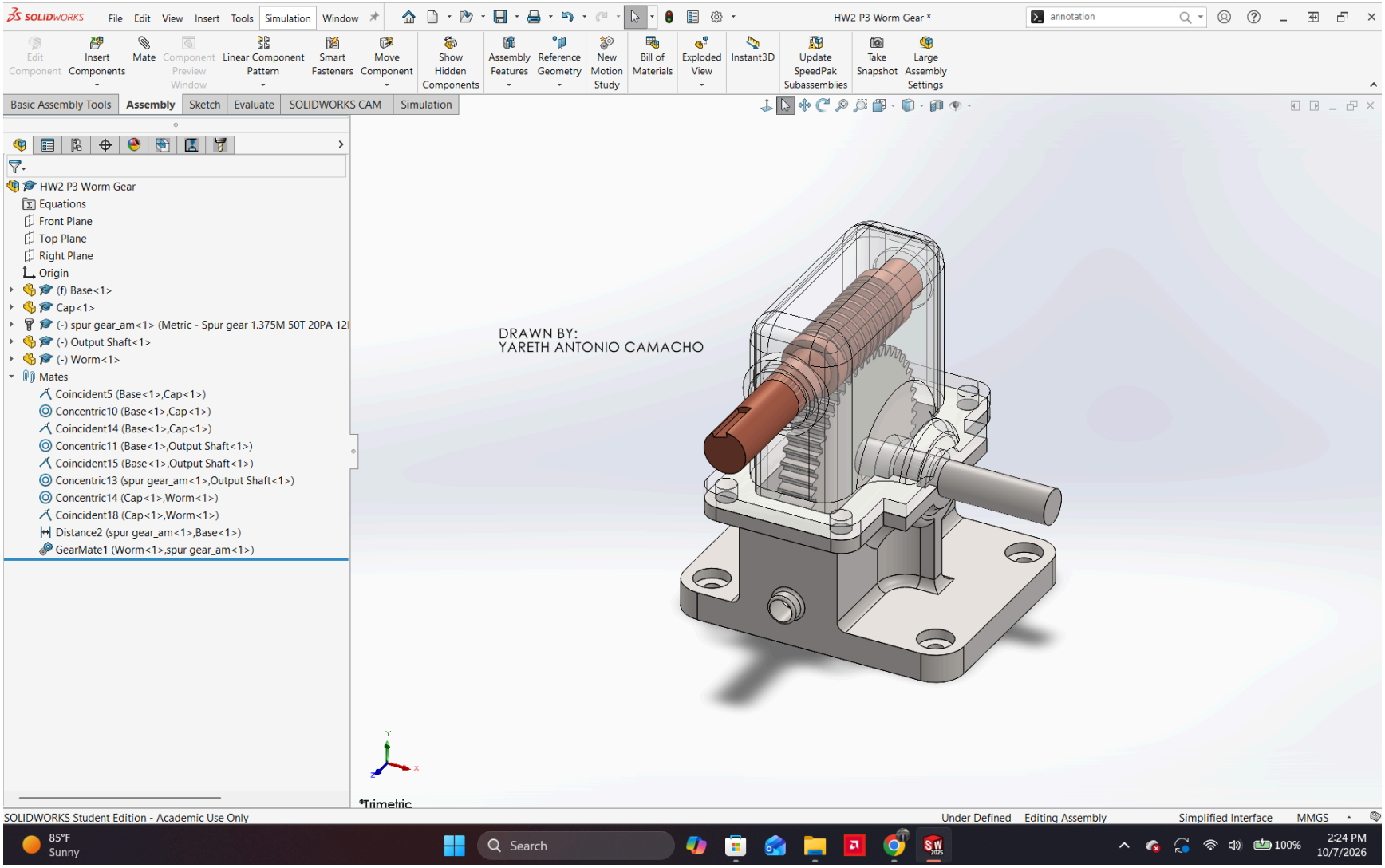
Homework Assignment 3 Assembly 2



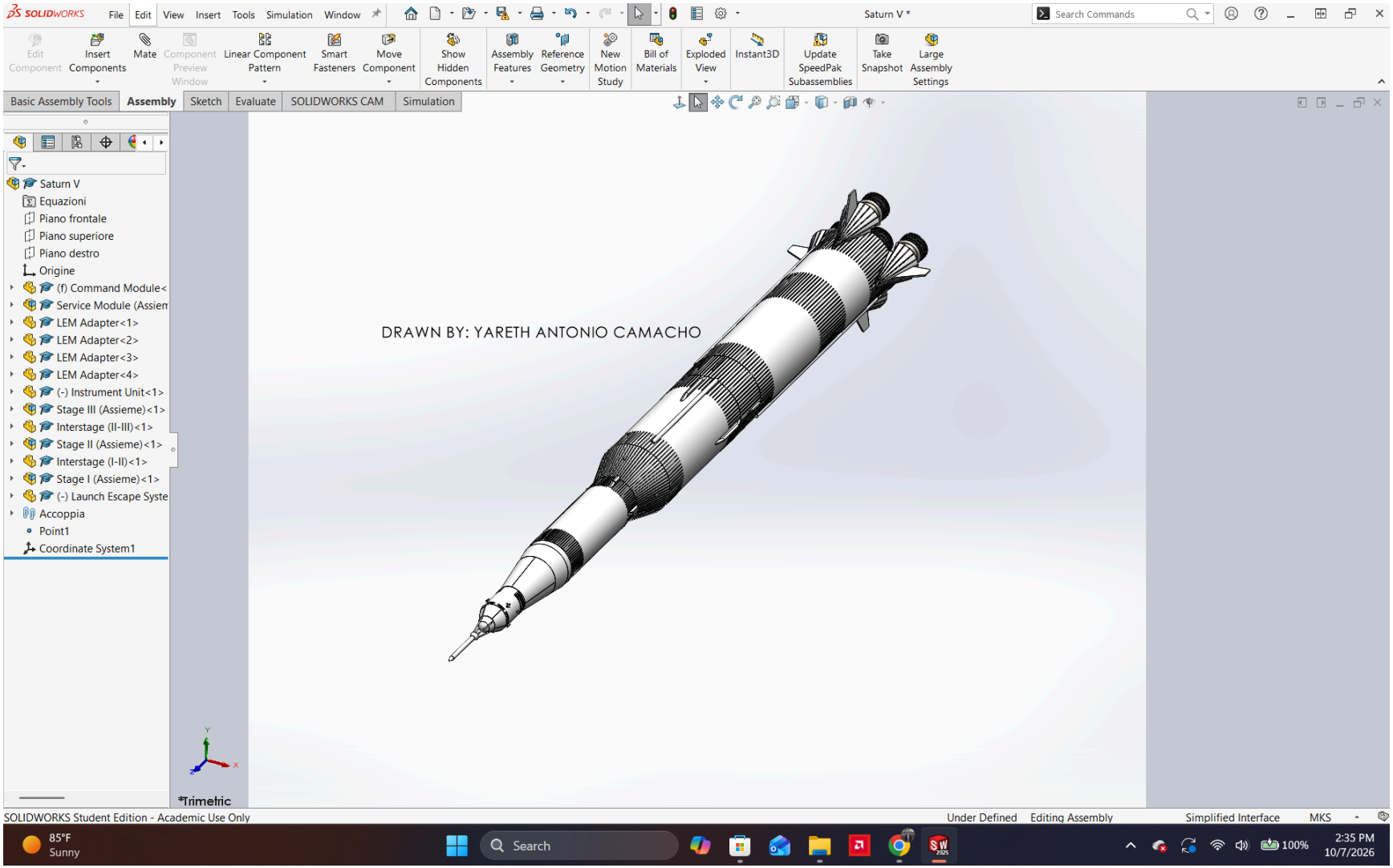
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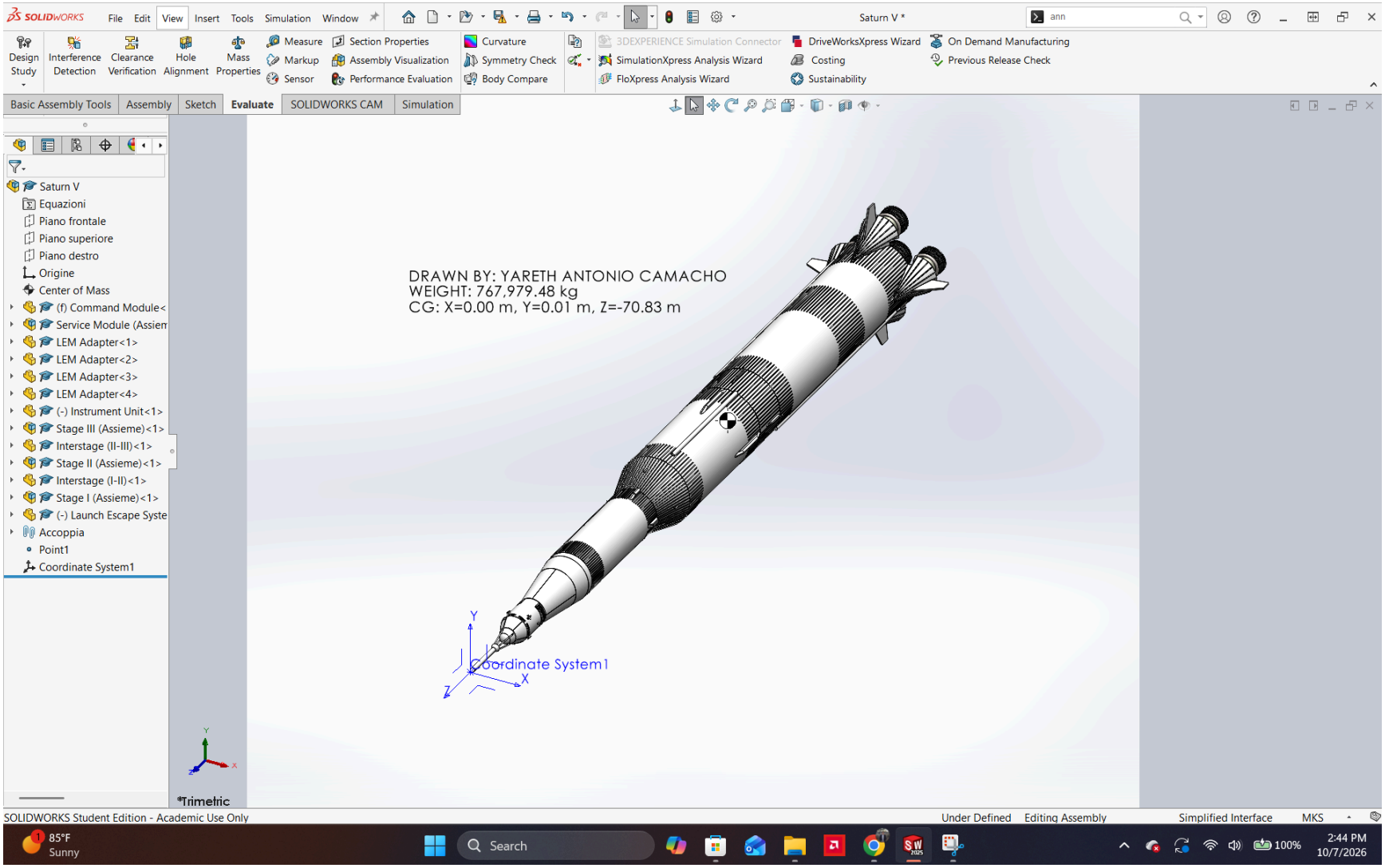
Homework Assignment 3 Assembly 4



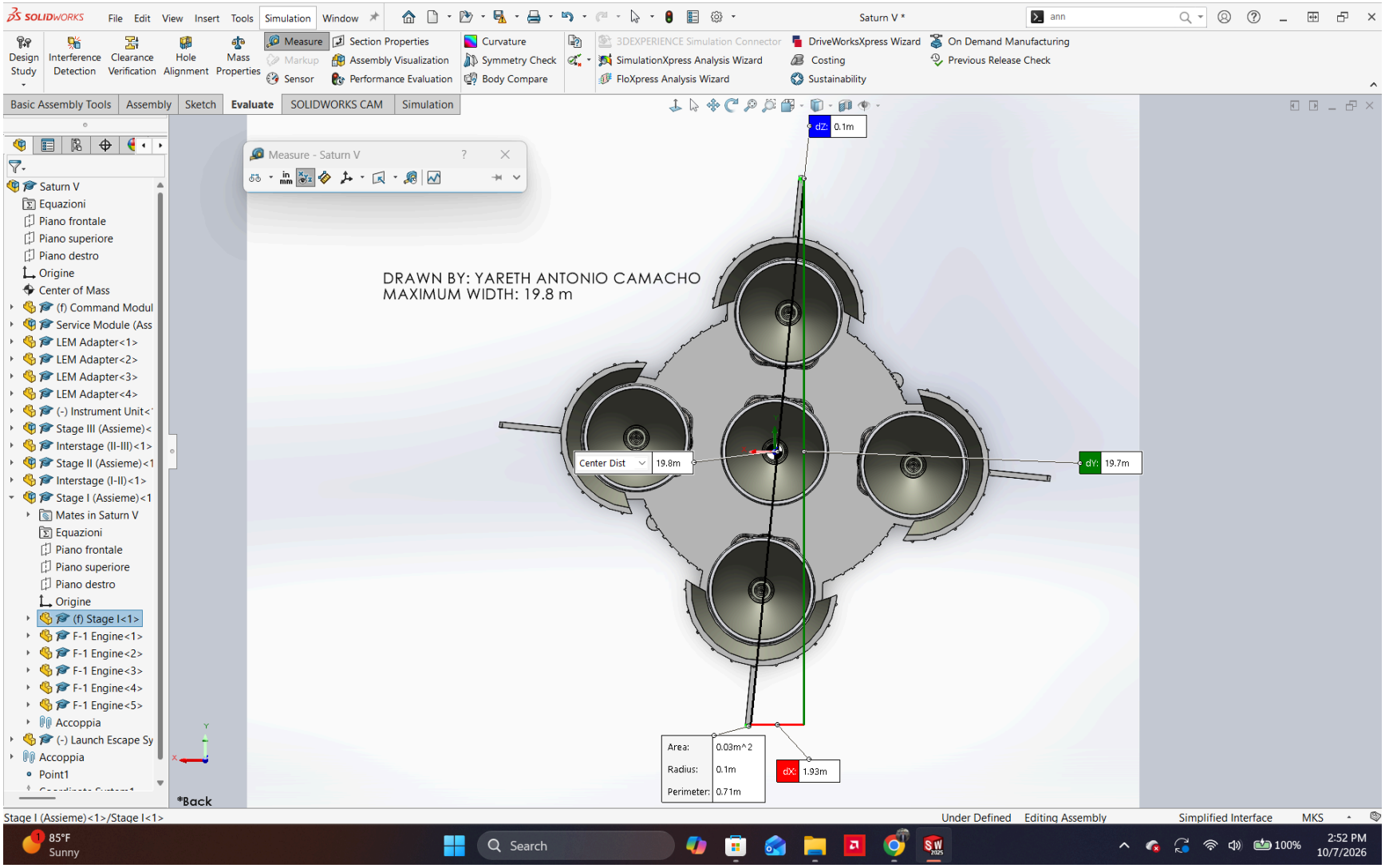
Homework Assignment 4 Part A



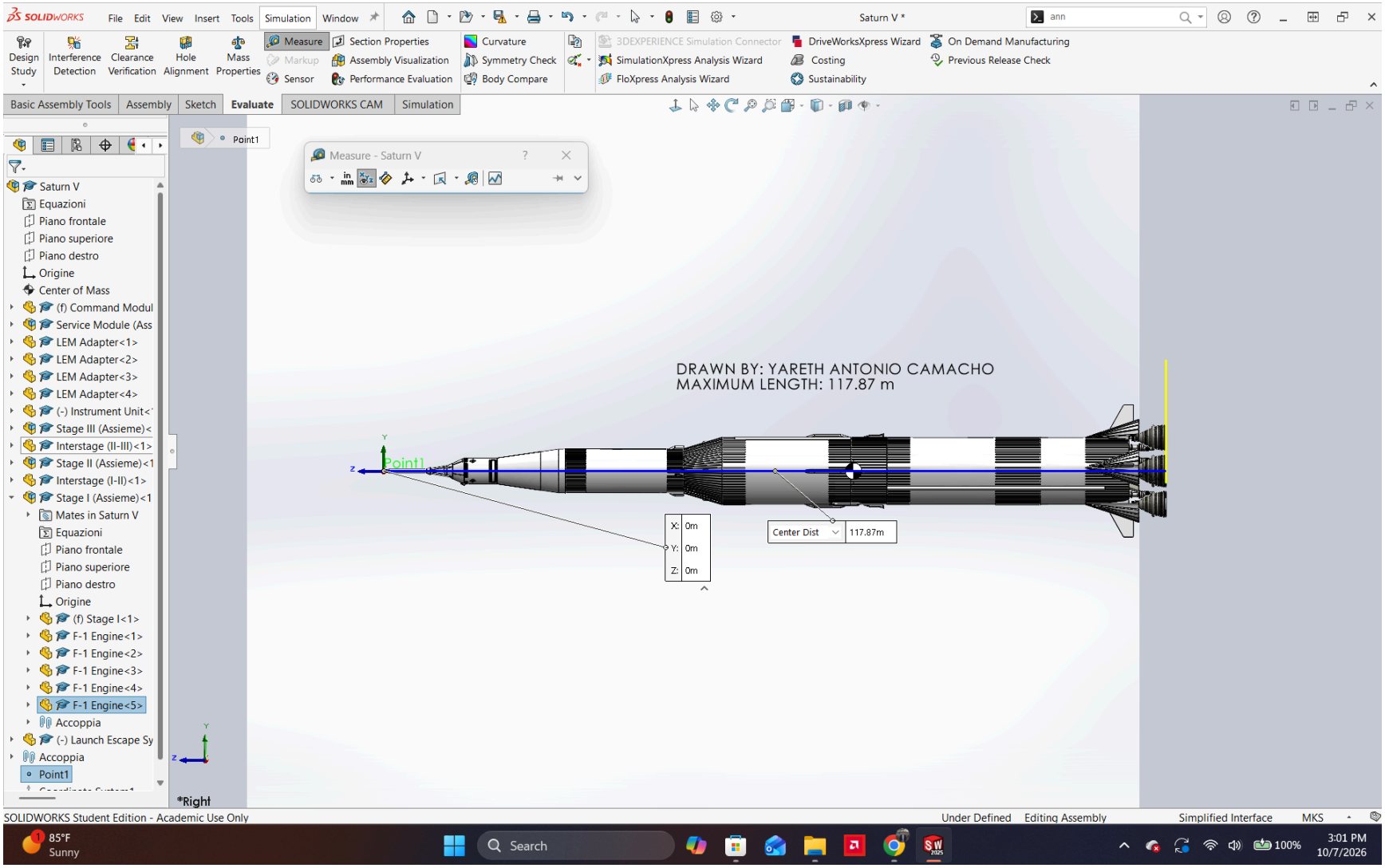
Homework Assignment 4 Part B



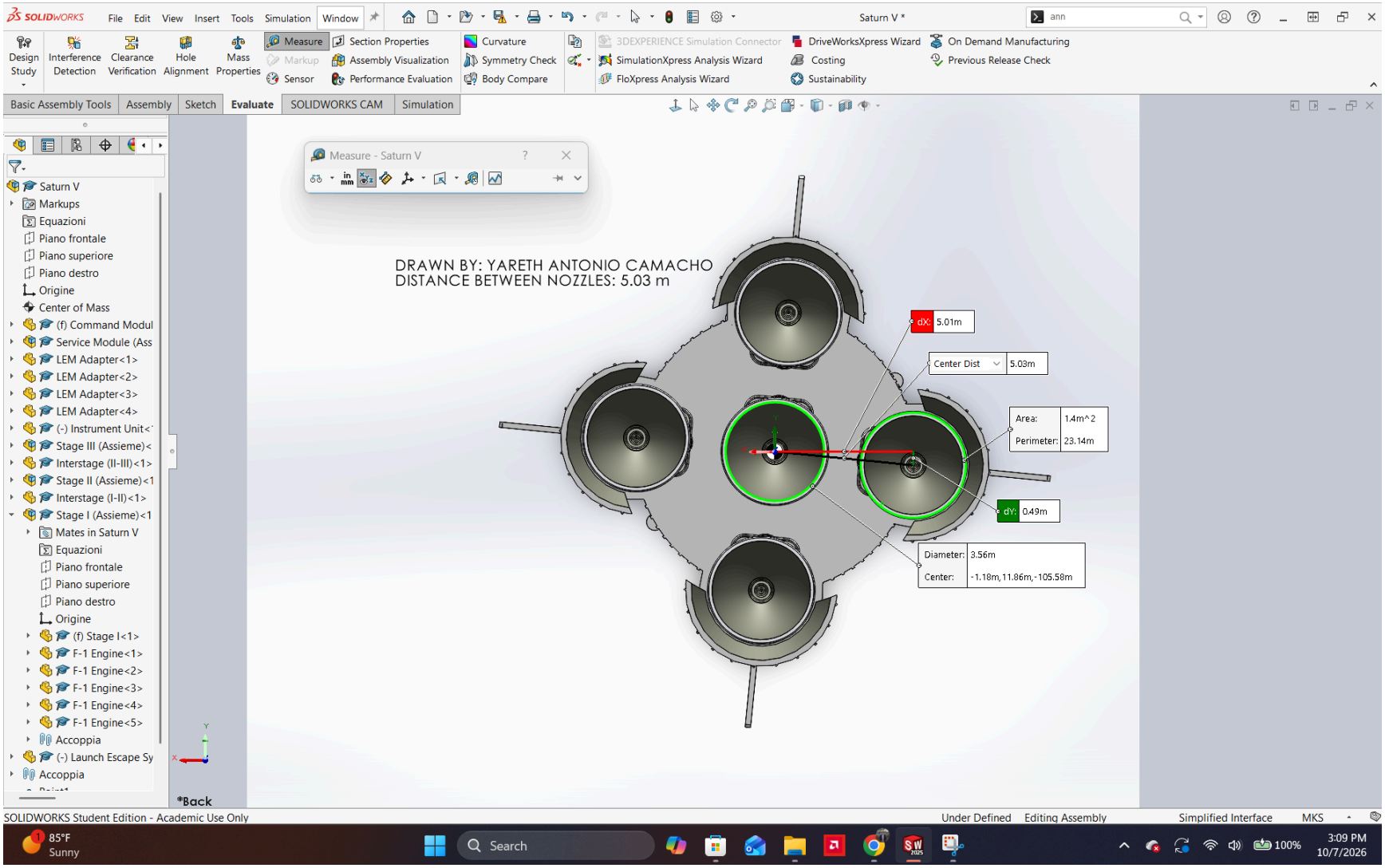
Homework Assignment 4 Part C



Homework Assignment 4 Part D

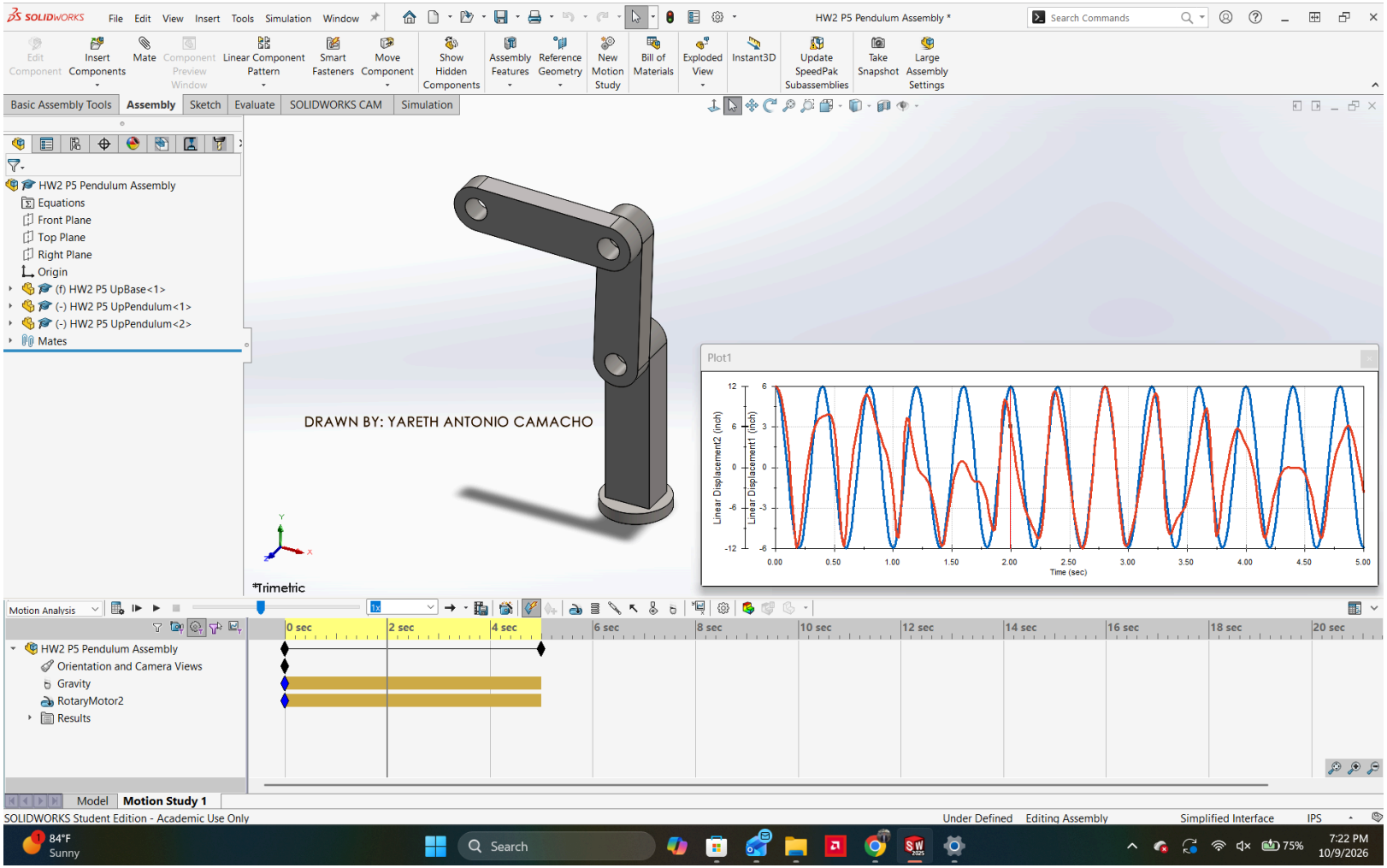


Homework Assignment 4 Part E

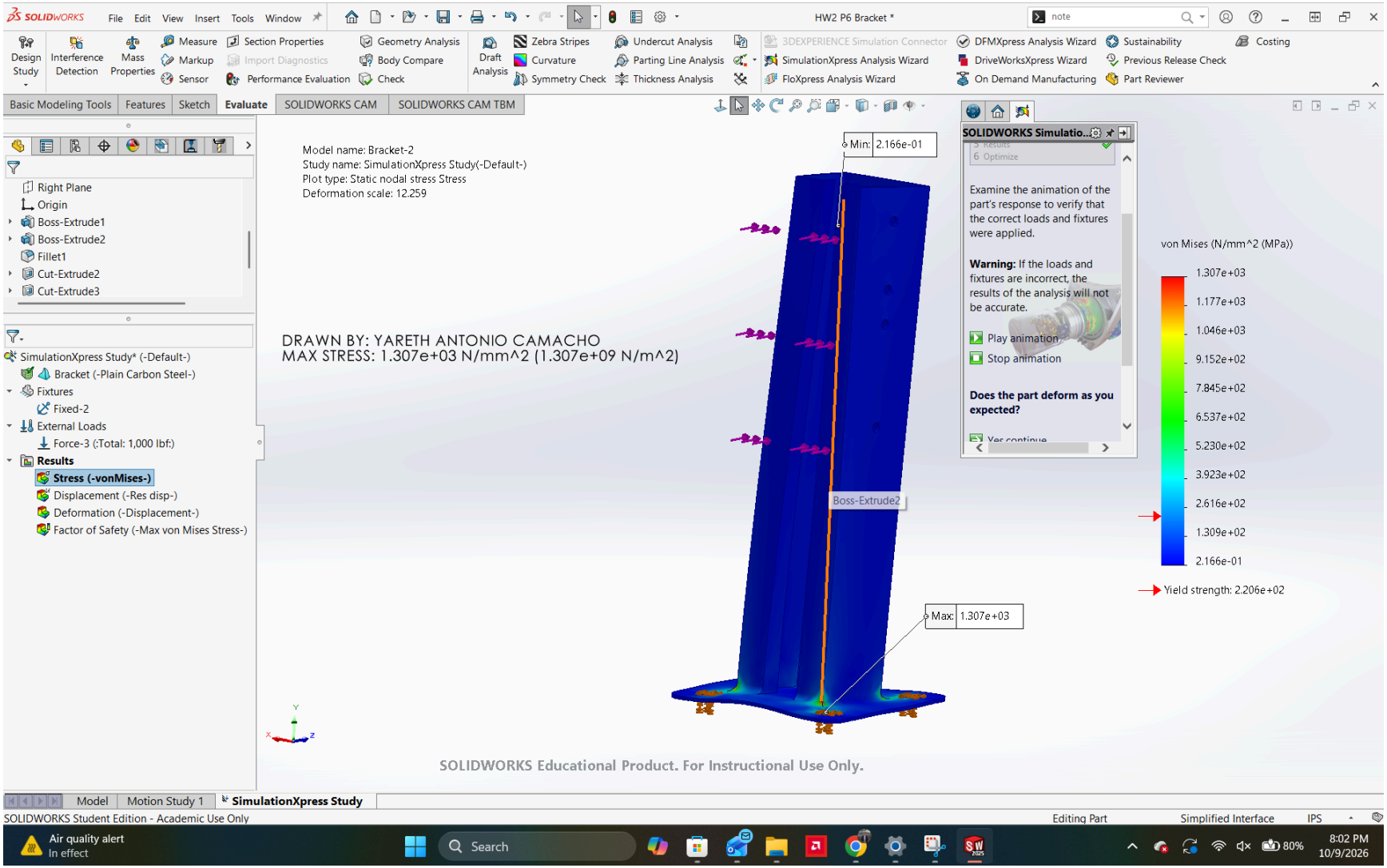


Homework Assignment 5

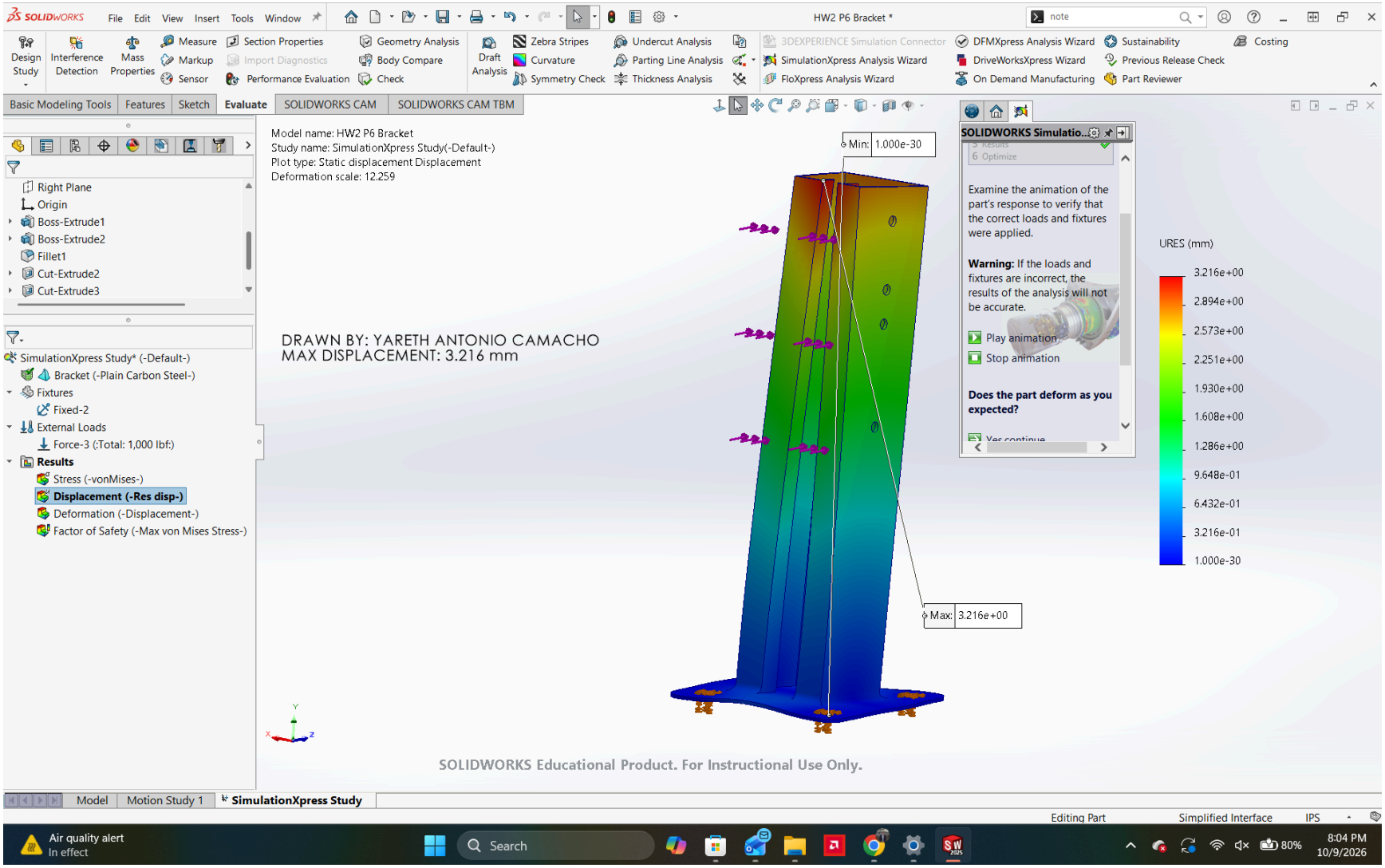
- The first Pendulum's plot is smooth because the motor spins at 150 RPM.
- The second pendulum's plot is uneven because nothing is driving it, it just swings freely under gravity



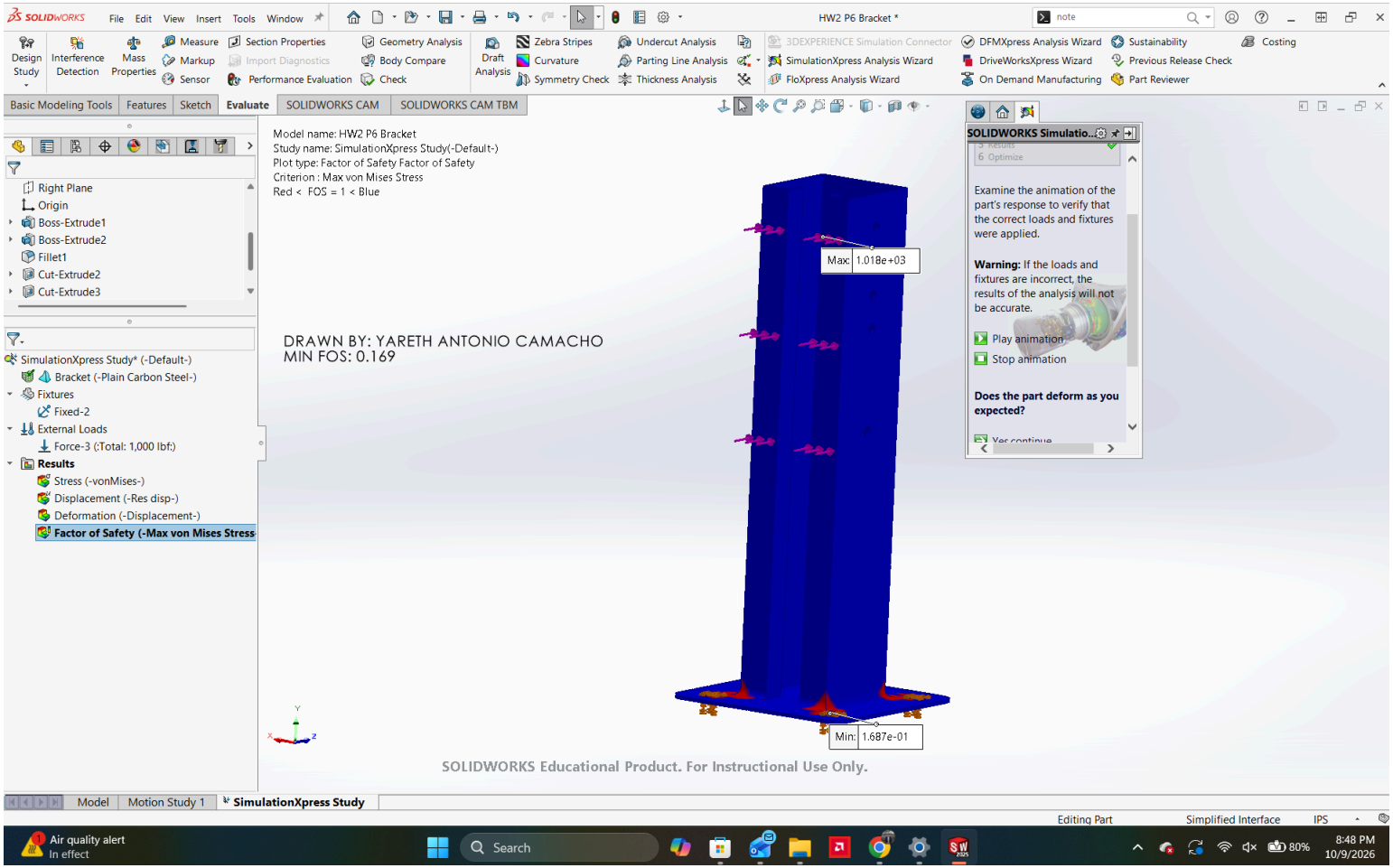
Homework Assignment 6 Part A



Homework Assignment 6 Part B



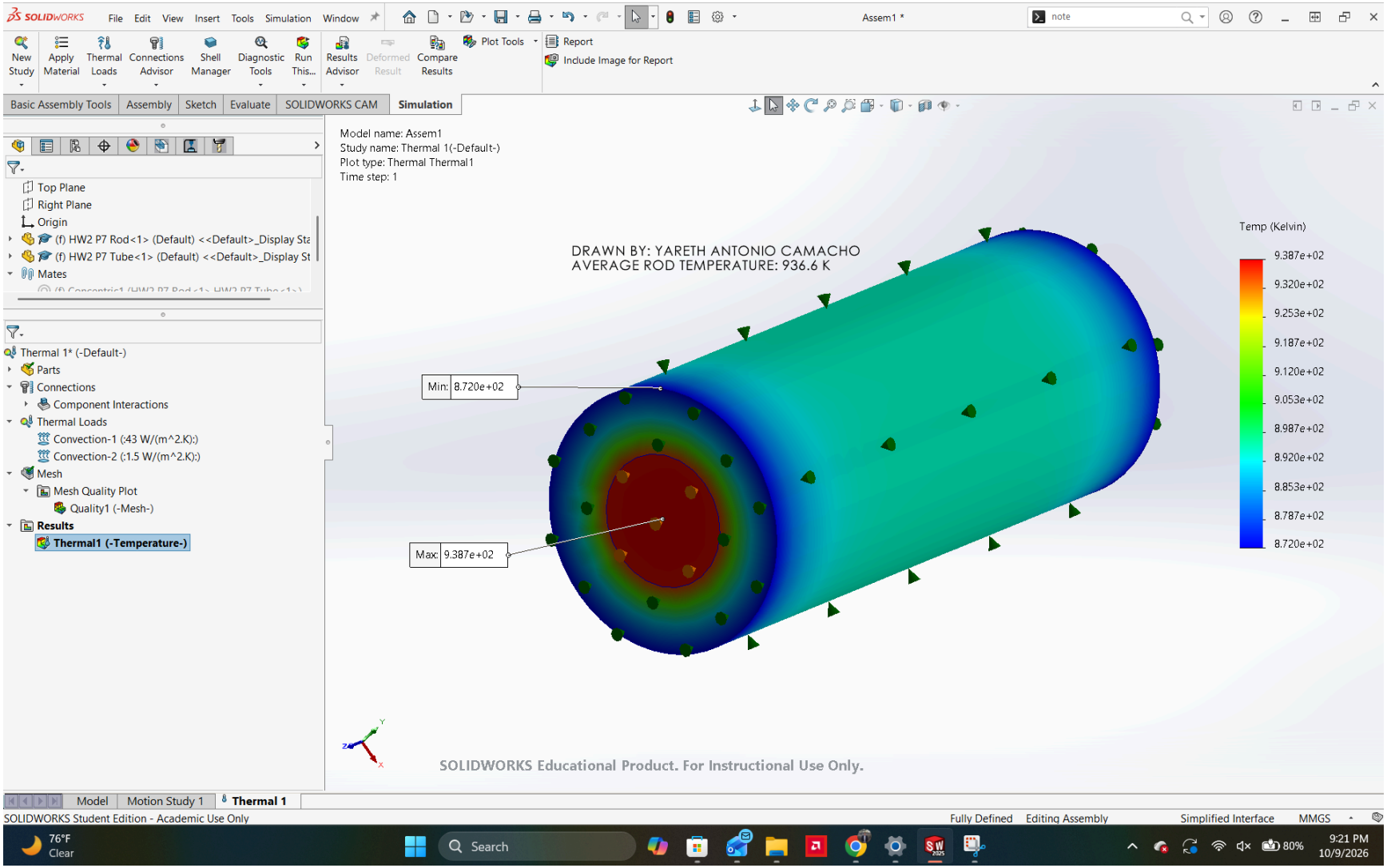
Homework Assignment 6 Part C



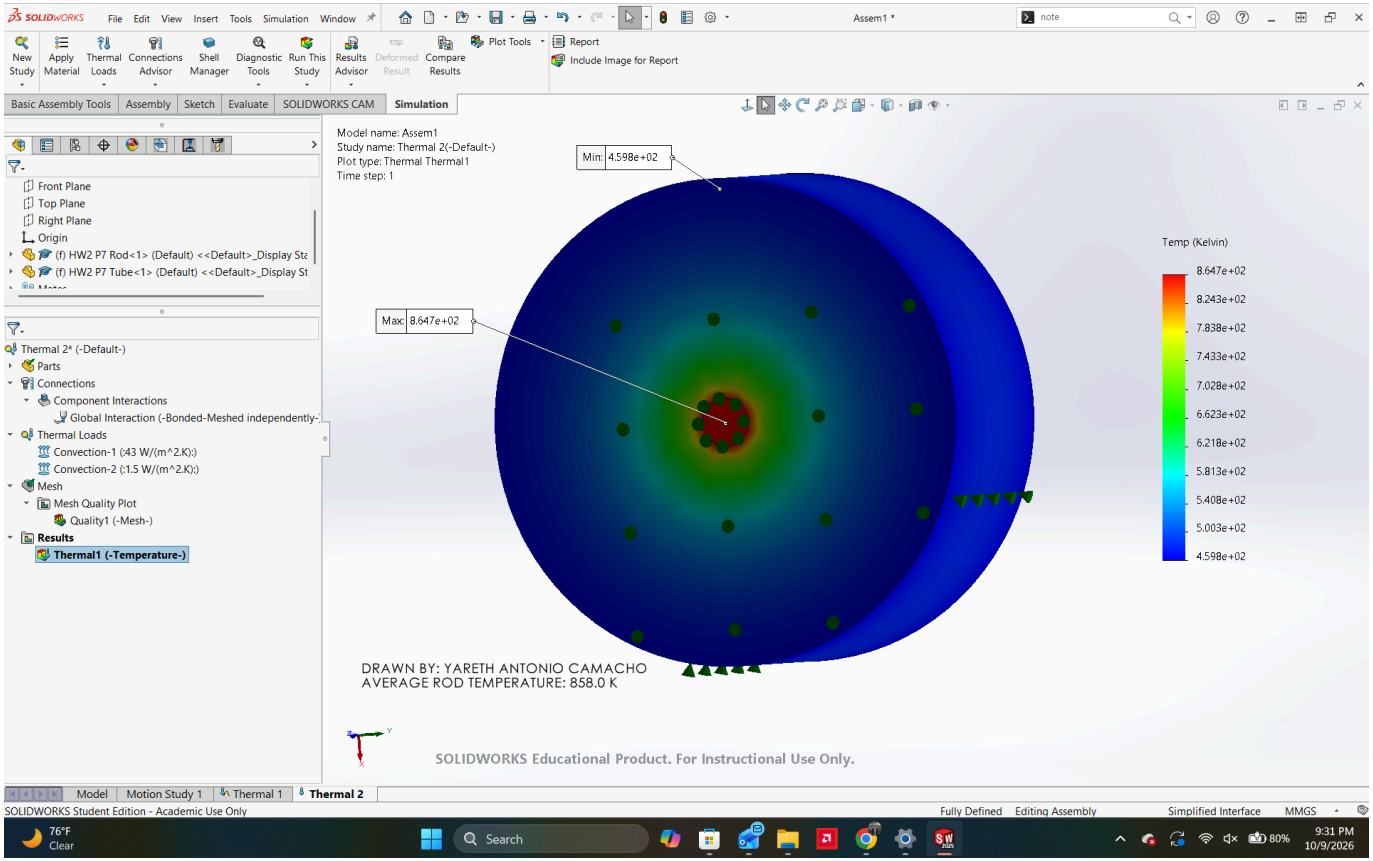
Homework Assignment 6 Part D

- The part fails at the base of the beam because the minimum factor of safety is 0.169 which is less than 1, so stress is higher than the steel's yield strength. It is expected because force pushes sideway on the beam so the bending occurs at the bottom.

Homework Assignment 7



Homework Assignment 8



- The average temperature for the rod decreased from 936.6 K in Problem 7 to 858.0 K in Problem 8. I expected it to stay hotter with the additional insulation added but the bigger tube has more surface touching the 300 K, so it loses more heat and results in the rod cooling down.